

Jimma University

Faculty of Computing and Informatics

Computer Science Department

Web Programming

Jimma, Ethiopia.

Web Design and Development Fundamentals

- Hypertext markup language (HTML)
- Why do we use HTML?
- HTML Document structure
- Basic HTML syntax
- Basic HTML tags

HTML PAGE STRUCTURE

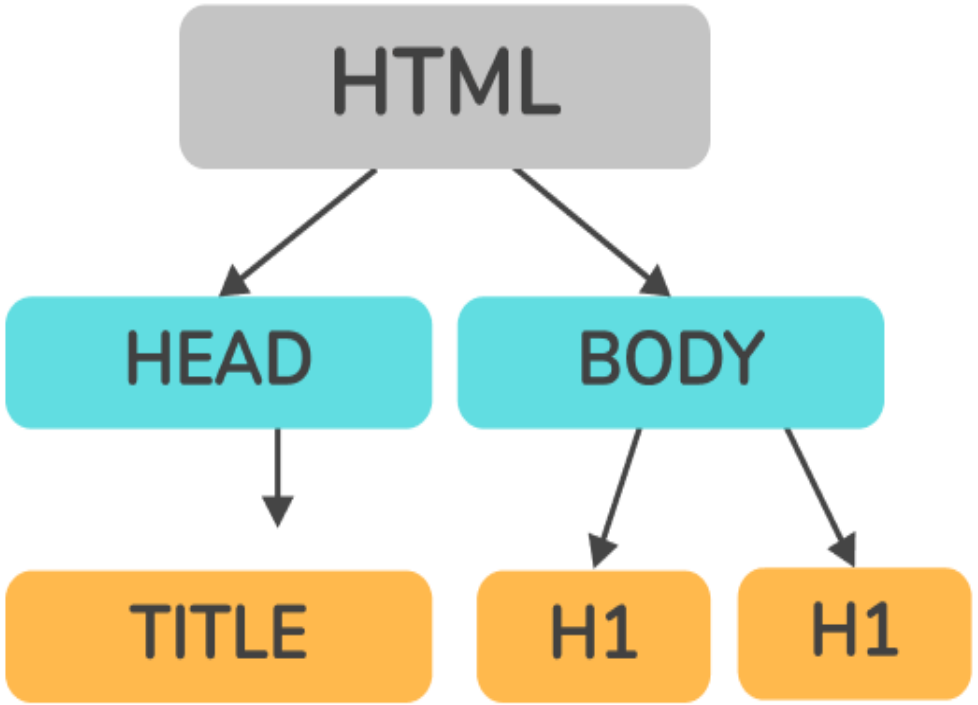


Introduction

- **HTML – HyperText Markup Language**—is the standard markup language **used to create web pages.**
- **HTML** describes the **structure of a website semantically,** along with cues for presentation
  - It is a markup language, rather than a programming language.
- **HTML** documents consist of a **tree of elements and text**.
  - Each element is denoted in the source by a start tag, such as **"<body>"**, and an end tag, such as **"</body>".**
- **Web browsers can read HTML files and render them into visible or audible web pages.**

Tree Structure

```
<!DOCTYPE html>
<html>
    <head>
        <title></title>
    <head>
    <body>
        <h1></h1>
        <h1></h1>
    <body>
 <html>
```



Introduction (cont'd)

- A markup language is a set of markup tags, and HTML **uses markup tags to describe web pages.**
- **HTML tags** are keywords surrounded by angle brackets like **<html>**
- **HTML tags** normally come in pairs like **** and ****.
  - The first tag in a pair is the **start tag**, the second tag is the **end tag**.
- **HTML elements** form the **building blocks of all websites.**
  - It allows **images and objects** to be embedded and can be used to create interactive forms.

Introduction (cont'd)

- HTML also provides a means to create structured documents by denoting structural semantics for text such as **headings, paragraphs, lists, links, quotes**, and **other items**.
  - It can **embed scripts in languages** such as **JavaScript** which **affect the behavior** of HTML webpages.
    - HTML can also be **used to include Cascading Style Sheets** (CSS) to define the **appearance** and **layout of text.**
- The purpose of a web browser is to read HTML documents and display them as web pages.
  - The browser does not display the HTML tags but uses the tags to interpret the content of the page

HTML, HEAD and BODY Tags

- The entire web page document is included within a **<html> tag**.
  - The <html> tag is called the **root element** because it contains all the elements in the document, and **it can not be** included within any other element.
    - Every web page **starts with <html> tag** and **ends with </html>**
- The HTML code is divided into two parts and both parts are **kept inside <html> tag**
  - The **first part is the head** and it starts with **<head>** and ends with **</head>.**
    - What we put in the head **will not be displayed to users.**
  - The **second part is the body** and it starts with **<body>** and ends with **</body> tag.**
    - Inside the <body> tag we **keep all the content** that we **want to display** to our web page users
    - **Whatever we place in** <body> will be displayed by the browser to the web users

HTML, HEAD and BODY Tags(cont'd)



HTML, HEAD, and BODY Tags(cont'd)
Anatomy of HTML document

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <title>Hello World</title>
  </head>
  <body>
    <h1>Hello World</h1>
    <p>This is a web page.</p>
  </body>
</html>
```

- The **<!DOCTYPE html>** declaration defines this document to be an HTML document.
- The **<html> element** is the root element of an HTML page
- The **lang attribute** defines the language of the document
- The **<meta> element** contains meta-information about the document
- The **charset attribute** defines the character set used in the document
- The **<title> element** specifies a title for the document
- The **<body> element** contains the visible page content
- The **<h1> element** defines a large heading
- The **<p> element** defines a paragraph

HTML, HEAD and BODY Tags(cont'd)

- <!DOCTYPE html>
- **<html></html>:** The <html> element.
  - This element wraps all the content on the entire page and it is known as the root element.
- **<head></head>:** The <head> element.
  - This element acts as a container for all the stuff you want to include on the HTML page that **isn't the content you are showing** to your page's viewers.
  - This includes things like **keywords** and a **page description** that you want to appear in search results, **CSS to style** content, **character set declarations**, and more.

HTML, HEAD and BODY Tags(cont'd)

- **<meta charset="utf-8">:**
  - This **element sets the character set your document should use to UTF-8**, which includes most characters from the vast majority of human written languages.
- **<title></title>:** The <title> element.
  - This sets the title of your page, which is the title that **appears in the browser tab** the page is loaded in
- **<body></body>:** The <body> element.
  - This contains **all the content that you want to show to web users** when they visit your page, whether that's text, images, videos, games, playable audio tracks, or whatever else.

HTML, HEAD and BODY Tags(cont'd)

<HEAD> … </HEAD>

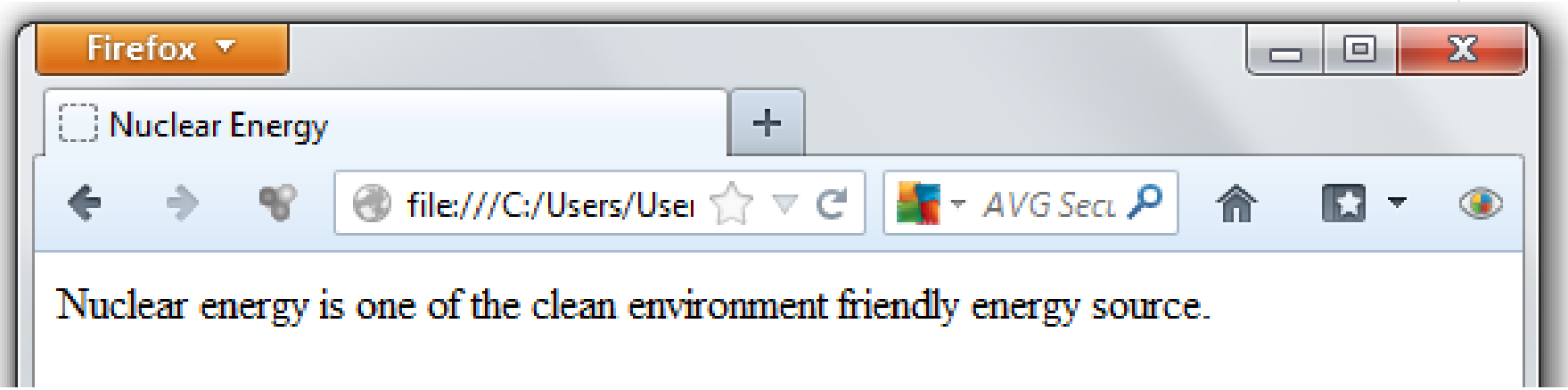
- The web page should have **only one head tag**. The head tag starts with <head> and ends with </head>.
- The text or tags that are inside the **head tag will not be displayed** in the browser window.
- **One of the important tags which is put inside <head> is <title> </title> tag.**
  - **Title tags are used to give a title to the browser window.**
  - Title tags are also important for our search engine point of view.
    - **We should keep the most important keywords inside the title tag.**

HTML, HEAD and BODY Tags(cont'd)

- It is also possible to put **JavaScript code and Cascading Style Sheets** in the head section.
  - If we are adding any JavaScript code here, then that will be **loaded when the browser opens the page.**

Example:

```
<html>
<head>
<title> Nuclear Energy </title>
</head>
<body>
   Nuclear energy is one of the clean environment-friendly energy sources.
</body>
</html>
```



HTML, HEAD and BODY Tags(cont'd)

The <title> Element

- You should specify a title for every page that you write inside the **<title> element**.
  - This element is a child of the **<head> element**.
- **It is used in several ways:**
  - It displays at the very top of a browser window.
  - It is used as the default name for a bookmark in browsers.
  - It is used by search engines that use its content to help index pages.
- Example: Here is an example of using a title tag.

```
<head>
<title>Nuclear Energy</title>
</head>
```

HTML, HEAD and BODY Tags(cont'd)

<BODY> … </BODY>

- This is where **we place our content for our visitors**. What we place here will be displayed to our visitors.
- A <body> element may contain anything from a couple of paragraphs under a heading to more complicated layouts containing forms and tables.

HTML, HEAD and BODY Tags(cont'd)

Attributes of HTML Tags

- HTML tags can have attributes. Attributes provide additional information about a tag.
- Tags often have attributes that modify them in some way. ***Attributes are always specified in the start tag.***
- Attributes come in **name-value pairs** like ***name="value"***
  - The syntax for attributes is as follows:

    <element attribute-name="value"> Content </element>
  - or **for empty elements**:

    <element attribute-name="value" />
  - **Example:** the **background color** of an **HTML document** can be changed using "bgcolor" attribute of the <body> tag.

    <body bgcolor="green">

HTML, HEAD and BODY Tags(cont'd)

<html> Tag Attributes

- The **lang attribute** of the HTML tag can be used to declare the language of a Web page or a portion of a Web page.
  - This is meant to assist search engines and browsers.
- According to the W3C recommendation, it is good to declare the primary language for each Web page with the **lang attribute inside the <html> tag**, like this:

```
<html lang="en">
  …
</html>
```

HTML, HEAD and BODY Tags(cont'd)

- ISO 639-1 defines abbreviations for languages.
- In HTML, they can be used in the lang attributes.

Language	ISO Code
Chinese	zh
English	en
French	fr
German	de
Russian	ru
Local language	
Afar	aa
Amharic	am
Afaan Oromo	om
Somali	so
Tigrinya	ti

HTML, HEAD and BODY Tags(cont'd)

```
<html lang="en">
<head>
    <title>Coffee</title>
</head>
<body>
  Jimma is the origin of Arabica coffee.
</body>
</html>
```

```
<html lang="om">
<head>
    <title>  Buna </title>
</head>
<body>
   Jimmi biyya bunaati.
</body>
</html>
```

HTML, HEAD and BODY Tags(cont'd)

<body> Tag Attributes

A. Background Color

- You can change the background color of your web page by using <BODY> tag attribute *bgcolor*
  - <body bgcolor="green"> and <body bgcolor="#00FF00">

B. Background Image

- We can use a background picture for the web page instead of a background color.
  - <body background="image1.gif">

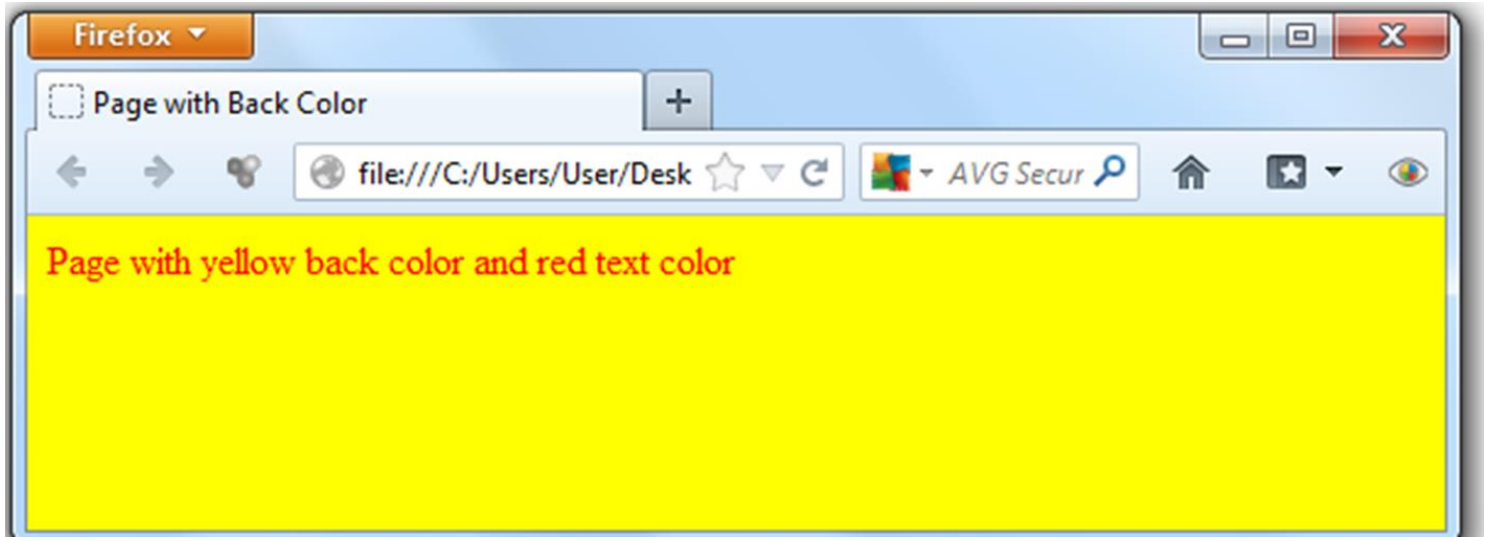
C. Text Color

- We can also set the text color of the web page just like the background color.
  - <body bgcolor="yellow" text="red">

HTML, HEAD and BODY Tags(cont'd)

Example:

```
<HTML>
<HEAD>
<TITLE>Page with Back Color</TITLE>
</HEAD>
<BODY bgcolor="yellow" text="#FF0000">
    Page with yellow back color and red text color
</BODY>
</HTML>
```



Creating Links

- A **hyperlink or link** is a word, group of words, or image that you can click on to jump to a new document or a new section within the current document.
- Web pages can contain links that take you directly to other pages or specific parts of the given page. **These links are known as hyperlinks.**
  - Hyperlinks allow visitors to navigate between Web sites by clicking on words, phrases, and images. Thus you can create hyperlinks **using text or images** available on your web page.
- Links are specified in HTML using the **<a> tag**. **This element is called an anchor tag** as well.
  - Anything between the opening <a> tag and the closing tag **becomes part of the link.**
- The **<a> tag** can be used in ***two ways:***
  - To **create a link to another document**, by using the **href attribute**
  - To **create a bookmark inside a document**, by using the **name attribute**

Creating Links (cont'd)

Link attributes: href, target, name, title, accesskey

The href Attribute:

- A link in HTML is always composed of two parts, the clickable part (the link text) and the URL (the destination site, page, or resource).
  - The URL is specified using the href attribute.
- The href (hypertext reference) attribute provides the address of the page (its URL) to the browser.
  - The URL must always appear in quotation marks.
- Here is an example:

```
<a href="http://ju.edu.et"> Jimma University</a>
```

Creating Links (cont'd)

The target attribute

- It is used to **specify where to display** the contents of a selected **hyperlink**.
- If set to:
  - ***_blank*** then a new window will be opened to display the loaded page
  - ***_top*** or ***_parent*** then the same window will be used to display the loaded document.
  - ***_self*** then loads the new page in the current window. By default, it's **_self**.
- Example: a link that opens on a new window/tab

```
<a href="first.html" target="_blank"> First Page</a><br>
```

Creating Links (cont'd)

The name Attribute

- The name attribute specifies the name of the anchor. The name attribute is **used to create a bookmark inside an HTML document.**
- Bookmarks are invisible to the reader.
  - Example: a named anchor inside an HTML document:

```
<a name="example">Sample Examples</a>
```

  - ➢ Then create a link to the "Sample Examples" inside the same document:

```
<a href="#example">Go to Sample Example</a>
```

The title attribute

- The title attribute is used to **type a short description of the link.**
  - If you place the cursor over the link, you will see the text in the title attribute.
- Example:

```
<a href="http://www.html.net/" title="Visit our website and learn HTML">Learn HTML </a>
```

Creating Links (cont'd)

The accesskey attribute:

- The accesskey attribute provides a **keyboard shortcut for the link**.
  - For example, you could make the T key an access key so that when the user presses either the Alt or Ctrl key on his keyboard along with the T key, the link gets activated.
- Example:

```
<a href="first.html" target="_blank" accesskey="T">First page</a>
```

Creating Links (cont'd)

Linking to email

- It is also possible to make a link to an e-mail address. It is done in almost the same way as when you link to a document.
  - To link to an email, you type ***mailto:*** followed by an e-mail address.
- Example:

```
<a href="mailto:home@gmail.com">Send an e-mail to us </a>
```

  - When the link is clicked, the **default e-mail program opens** with a new blank message addressed to the specified e-mail address.
    - ➢ This function will only work if there is an e-mail program installed on your computer.
- You can **make an image a link** that can be clicked to open a new page.
  - To make an image a link, simply put the img element in the anchor element:

```
<a href="http://www.oreilly.com"><img src="ora.gif" /></a>
```

Creating Links (cont'd)

❖ There are two ways to specify the URL:

1. Absolute URL: Absolute URLs **provide the full URL for the document**, including the protocol, the domain name, and the pathname as necessary.

- You need to use an absolute URL when pointing to a document out on the Web.
  - ➢Example: Jimma university

2. Relative URL: Relative URLs describe the **pathname to the linked file relative to the current document.**

- It **doesn't require the protocol** or domain name—just the pathname.
- Relative URLs can be used when you are linking to another document **on your own site** (i.e., on the same server).
  - ➢Example:

** Recipes **

HTML Text Formatting Tags

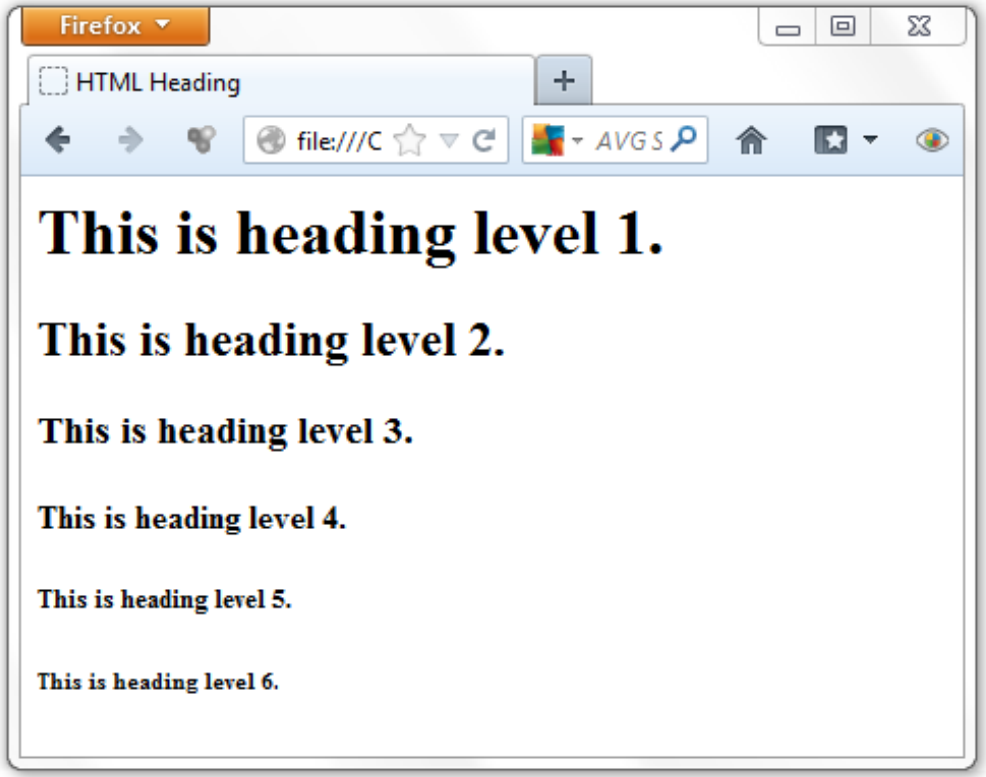
- HTML Headings
- HTML paragraph
- HTML Font—face, type, color
- HTML bold <b>, italicized <i>, or underlined <u>
- Line Breaks - the
 Element
- Preserve Formatting - the <pre> Element
- Horizontal Rules - the <hr> Element
- Subscript <sub> and Superscript <sup>
- Special Characters

HTML Text Formatting Tags

HTML Headings

- In HTML, you can **create different heading levels in your document** to help you organize the document into sections, just as you might do when writing a book.
- A heading element briefly describes the topic of the section it introduces.
- Headings are defined with the **<h1> to <h6> tags**.
  - <h1> defines the largest heading and
  - <h6> the smallest heading.

```
<body>
  <h1>This is heading level 1. </h1>
  <h2>This is heading level 2. </h2>
  <h3>This is heading level 3. </h3>
  <h4>This is heading level 4. </h4>
  <h5>This is heading level 5. </h5>
  <h6>This is heading level 6. </h6>
</body>
```



HTML Text Formatting Tags (cont'd)

HTML Paragraphs

- Paragraphs are defined with the **<p> tag.**
- Example:

```
<p>This is a paragraph</p>

<p>This is another paragraph</p>
```

- **Both of the above tags will enforce a new line whenever you write the tag.**

HTML Text Formatting Tags (cont'd)

The align attribute

- You can use the **align attribute to align your paragraphs**.
  - Paragraphs can be aligned left, center, right, or justified. You can do this by using the align attribute.
- Align attributes can be used also with other tags like headers, etc.
- Example:

```
<p align="center">This is center aligned.</p>

<p align="justify">This is justified. This works when you have multiple lines in your paragraph and you want to justify all the lines so that they look nicer.</p>
```

HTML Text Formatting Tags (cont'd)

HTML Fonts

- The ** tag** is used to add **font type, size, and color** to the text on your site.
- The font tag has three attributes
  - **size,**
  - **color, and**
  - **face**
- They are used to customize your fonts.
- To change any of the font attributes at any time within your page, simply use **the tag.** The text that follows will remain changed until you close with the tag.
- You can change any or all of font attributes at the one time.

HTML Text Formatting Tags (cont'd)

Font Size:

- You can set the size of your font with the size attribute.
  - The range of accepted values is from **1(smallest) to 7(largest).**
- The default size of a font is 3.
  - Example:

```
<font size="1">Font size=1</font>
<font size="2">Font size=2</font>
```

Specify the Relative Font Size

- You can also specify **relative font sizes instead of exact font sizes.**
- This can be done like:

```
<font size="+n">
<font size="-n">
```

- This specifies how many sizes larger or how many sizes smaller than the **preset font size should be.**

Example:

```
<font size="-1">Font size="-1"</font>
<font size="+1">Font size="+1"</font>
<font size="+2">Font size="+2"</font>
<font size="+3">Font size="+3"</font>
<font size="+4">Font size="+4"</font>
```

HTML Text Formatting Tags (cont'd)

Font Face:

- You can set any font you like **using the *face* attribute**
  - But be aware that if the user viewing the **page doesn't have the font installed,** they will not be able to see it.
    - Instead they will default to Times New Roman of your font with size attribute.
- Example:

```
<font face="Times New Roman" size="5"> Times New Roman </font>
<font face="Verdana" size="5"> Verdana </font>
<font face="Lucida Calligraphy, Comic Sans MS, Lucida Console”>
```

HTML Text Formatting Tags (cont'd)

Font Color:

- You can set any font color you like using a color attribute.
- You can specify the color that you want by either the **color name** or **hexadecimal code** for that color. Check a complete list of HTML Color Name with Codes.
- Example:

```
<font color="#0000FF">This text blue color</font>
<font color="red">This text is red</font>
```

HTML Text Formatting Tags (cont'd)

Formatting Tags

- HTML provides many tags that you can **use to enhance and change the look of the text.**
  - You can make **text bold, italicized, or underlined**.
- These are just some of the presentational options available to indicate how text can appear in HTML.

Bold Text - the Element:

- Anything that appears in a ... element is displayed in bold.
- The tag makes text to be displayed in boldface.

Example:

```
<p>The use of nuclear energy needs <b> safety caution </b> because of the associated danger.</p>
```

Output:

The use of nuclear energy needs **safety caution** because of the associated danger.

HTML Text Formatting Tags (cont'd)

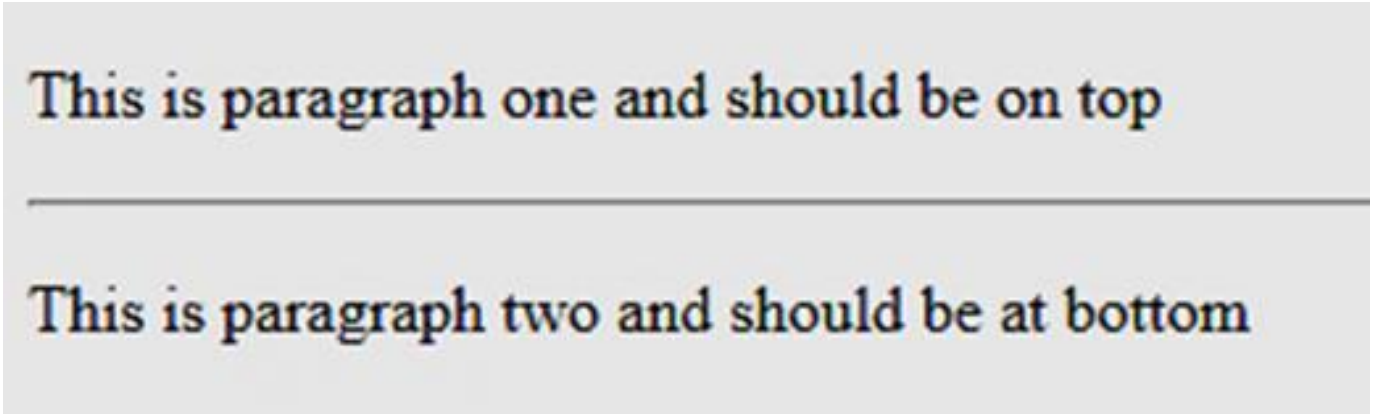
- **Italic Text - the Element:** *italicized*
- **Underlined Text - the <u> Element:** <u>underlined</u>
- **Create Line Breaks - the
 Element:** The
 tag inserts a **single line break**. You can also use **
** which does the same.
- **Preserve Formatting - the <pre> Element:** Sometimes you want your **text to follow the exact format of how it is written in the HTML document**.
  - In those cases, you can use the preformatted tag (<pre>).
  - The <pre> tag defines preformatted text.
    - Text in a pre-element is displayed in a fixed-width font (usually Courier),
      - It preserves **both spaces and line breaks**.

HTML Text Formatting Tags (cont'd)

Horizontal Rules - the <hr> Element

- <hr> Horizontal rules are used to **visually break up sections of a document**.
- The <hr> tag creates **a horizontal line in an HTML page**.

```
<p>This is paragraph one and should be on top</p>
<hr>
<p>This is paragraph two and should be at the bottom</p>
```



HTML Text Formatting Tags (cont'd)

Subscript and Superscript Text:

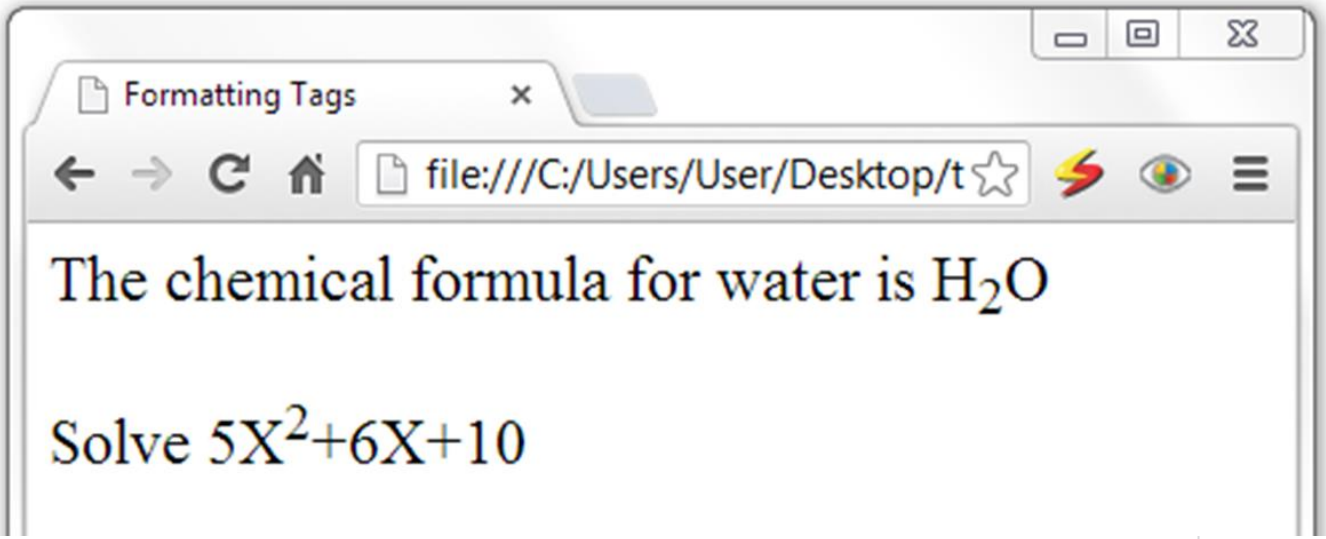
- The <sub> tag defines subscript text.
  - Subscript text **appears half a character below the baseline.**
  - Used for chemical formulas, like H_2O.
- The <sup> tag defines superscript text.
  - Superscript text **appears half a character above the baseline.**
  - Used for mathematical expressions like x^2+y or footnotes, like HTTP[1].
- Example:

```
<p> The chemical formula for water is H<sub>2</sub>O </p>

<p> Solve 5X<sup>2</sup>+6X+10 </p>
```

HTML Text Formatting Tags (cont'd)

```
<html>
<head> <title>Formatting Tags</title> </head>
<body>
<font size="5">
<p> The chemical formula for water is H<sub>2</sub>O </p>
<p> Solve 5X<sup>2</sup>+6X+10 </p>
</font>
</body>
</html>
```

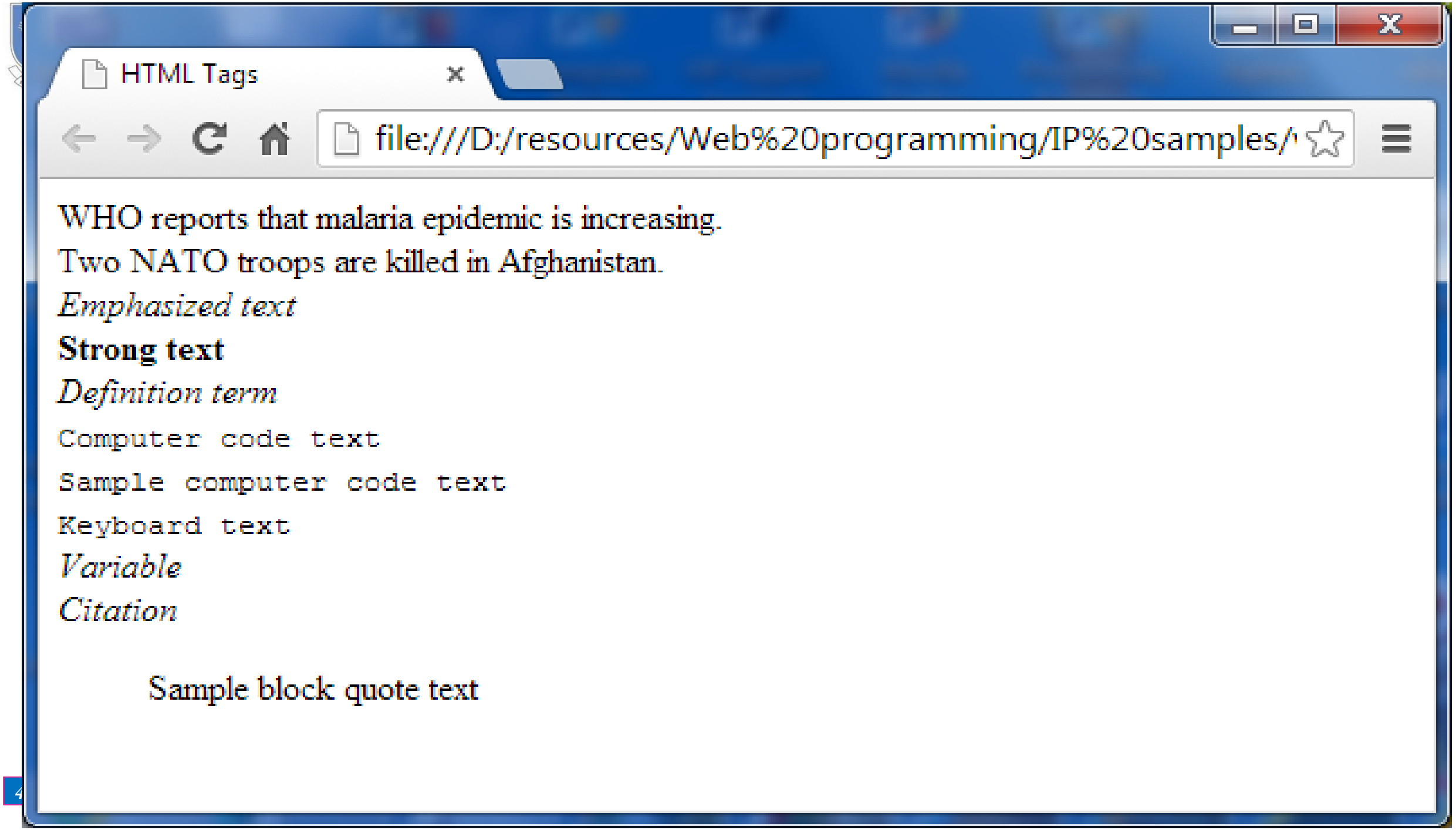


HTML Text Formatting Tags (cont'd)

Tag	Description
****	Renders as emphasized text
****	Renders as strong (highlighted) text
<dfn>	Defines a definition term
<code>	Defines computer code text
<samp>	Defines sample output from computer code
<kbd>	Defines keyboard text
<var>	Defines a variable part of a text
<abbr>	Defines an abbreviation
<q>	Defines a short quotation.
<blockquote>	To quote a passage from another source
<cite>	Defines a citation

Example of formatting

```
<html>
<head> <title>HTML Tags</title>
</head>
<body>
<abbr title="World Health Organization">WHO</abbr> reports that malaria epidemic is increasing. <br>
Two <acronym title="North Atlantic Treaty Organization">
NATO</acronym> troops are killed in Afghanistan. <br>
<em>Emphasized text</em> <br>
<strong>Strong text</strong> <br>
<dfn>Definition term</dfn> <br>
<code>Computer code text</code> <br>
<samp>Sample computer code text</samp> <br>
<kbd>Keyboard text</kbd> <br>
<var>Variable</var> <br>
<cite>Citation</cite>  <br>
<blockquote>Sample block quote text</blockquote>
</body>
</html>
```


HTML Tags
file:///D:/resources/Web%20programming/IP%20samples/
WHO reports that malaria epidemic is increasing.
Two NATO troops are killed in Afghanistan.
Emphasized text
Strong text
Definition term
Computer code text
Sample computer code text
Keyboard text
Variable
Citation
Sample block quote text

HTML Text Formatting Tags (cont'd)

Special Characters

- Characters within HTML documents that are **not part of a tag are rendered as-is by the browser.**
- However, **some characters have special meaning** and **are not directly rendered**, while other **characters can't be typed into the source document from a conventional keyboard.**
  - Special characters need either a **special name or a numeric character code** for inclusion in an HTML document.

Character	Code	Short name	Character name
"	"	"	quotation mark
'	'	'	apostrophe
&	&	&	ampersand
<	<	<	less-than
>	>	>	greater-than
			non-breaking space
¡	¡	¡	inverted exclamation mark
¢	¢	¢	cent
£	£	£	pound
¤	¤	¤	currency
¥	¥	¥	yen
¦	¦	¦	broken vertical bar
§	§	§	section
¨	¨	¨	spacing diaeresis
©	©	©	copyright
ª	ª	ª	feminine ordinal indicator
«	«	«	angle quotation mark (left)
¼	¼	¼	fraction 1/4
½	½	½	fraction 1/2
¾	¾	¾	fraction 3/4
¿	¿	¿	inverted question mark
×	×	×	multiplication
÷	÷	÷	division

¬	¬	¬	negation
			soft hyphen
®	®	®	registered trademark
¯	¯	¯	spacing macron
°	°	°	degree
±	±	±	plus-or-minus
²	²	²	superscript 2
³	³	³	superscript 3
´	´	´	spacing acute
µ	µ	µ	micro

¶	¶	¶	paragraph
·	·	·	middle dot
¸	¸	¸	spacing cedilla
¹	¹	¹	superscript 1
º	º	º	masculine ordinal indicator
»	»	»	angle quotation mark (right)
¼	¼	¼	fraction 1/4
½	½	½	fraction 1/2
¾	¾	¾	fraction 3/4
¿	¿	¿	inverted question mark
×	×	×	multiplication
÷	÷	÷	division

HTML Comments

- Comments are a piece of code that is **ignored by any web browser.**
  - It is good practice to comment on your code, especially in complex documents, to indicate sections of a document and any other notes to anyone looking at the code.
- Comments help you and others understand your code.
- HTML comment are indicated by **beginning tag <!--** and **ending tag -->**
- You can comment multiple lines by the special beginning tag <!-- and ending tag --> placed before the first line and end of the last line to be treated as a comment.
- For example Given line is a valid comment in HTML

<!-- This is commented out -->

Inserting Image

- Images are very important to depict many concepts on your web page or to make it visually appealing.
  - It is often said that a single image is worth thousands of words.
- In HTML, images are defined with **the <img> tag.**
  - The <img> tag has attributes, **but not a closing tag.**

<img> Tag Attributes

- **The src Attribute**
  - To display an image on a page, you need to use **the src attribute**.
  - The value of the **src attribute** is the **URL of the image you want to display.**
  - Syntax for inserting an image:

```
<img src="url" alt="Alternative Text"/>
```

Attribute	Description
src	The value of the src attribute is the URL of the image you want to display.
width	sets the width of the image.
height	sets the height of the image
alt	The alt attribute specifies an alternate text for an image if the image cannot be displayed
border	The border attribute sets a border around the image
align	The align attribute sets the horizontal alignment of the image and takes value either left, right, or center.
valign	The valign attribute sets the vertical alignment of the image and takes value either top, bottom or center.
hspace	sets horizontal space around the image
vspace	sets vertical space around the image
title	The title attribute specifies a text title

Inserting Image (cont'd)

- **The alt attribute**
  - The alt attribute specifies an **alternate text for an image** if the image cannot be displayed.
  - Example:

```
<img src="boat.gif" alt="Big Boat" />
```

- **The width and height attributes**
  - To define the **height and width of the image**, rather than letting the browser compute the size, use the height and width attributes.
  - width: sets the width of the image.
    - This can have a value like 10 (pixels) or 20% (percentage of the available window size) etc.
  - height: sets the height of the image.
    - This can have a value like 10 (pixels) or 20% (percentage of the available window size) etc.
- Example:

```
<img src="sunset.jpg" height="50" width="100">
```

Inserting Image (cont'd)

❖ **The border attribute**

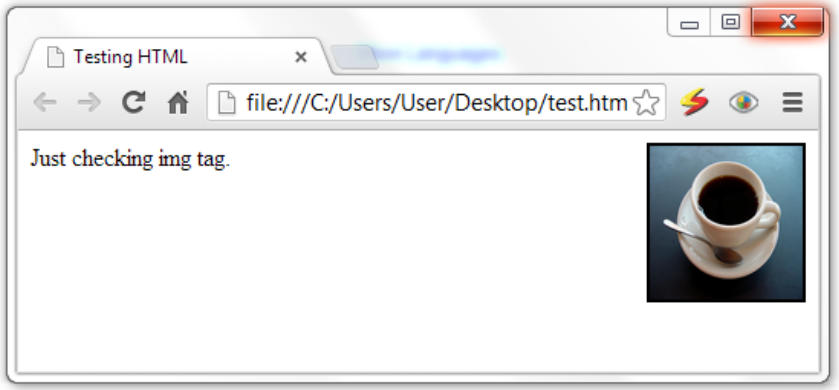
- The border attribute sets a border around the image. This will have a value like 1, 2, etc.

❖ **The align attribute**

- The align attribute **sets the horizontal alignment** of the image and takes value either **left, right, or center**.

❖ Example:

```
<img src="coffee.gif" alt="Highland coffee" width="100" height="100"
 border="2" align="right" title="Coffee products" />
```



Inserting Image (cont'd)

❖ The valign attribute

- The valign attribute **sets the vertical alignment of the image** and takes value either **top, bottom or center.**

❖ The hspace and vspace attributes

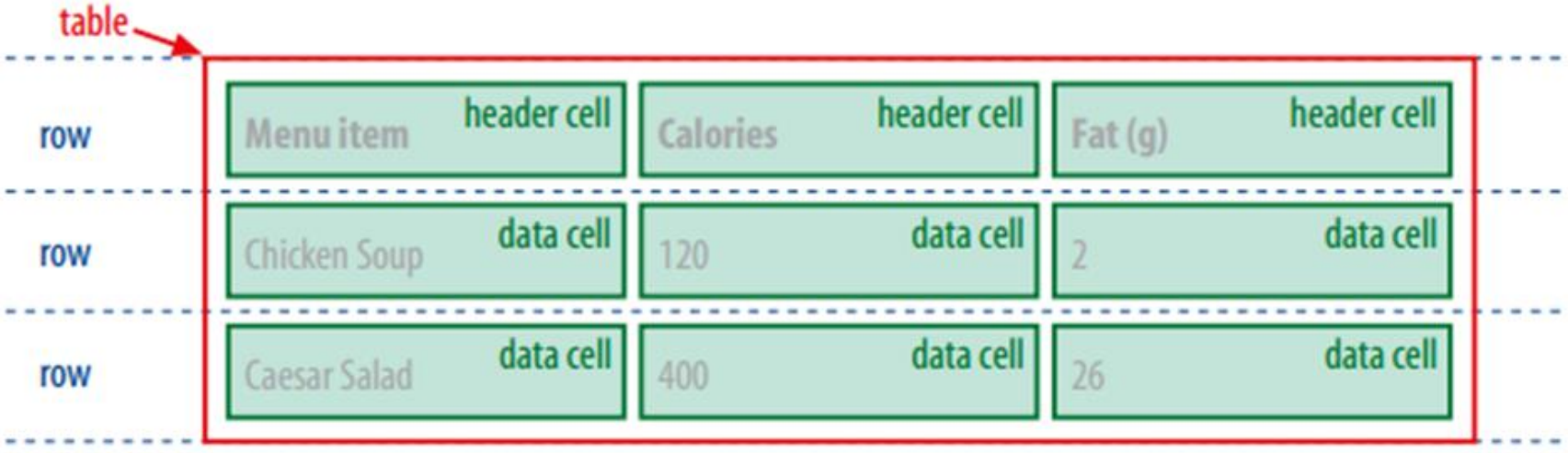
- **hspace:** sets horizontal space around the image.
  - ➢This will have a value like 10 or 20% etc.
- **vspace:** sets vertical space around the image.
  - ➢This will have a value like 10 or 20% etc.

❖ The title attribute

- The title attribute **specifies a text title.**
- The browser displays the title when the **mouse passes over the link.**

Tables

- Tables are defined with the ***<table> tag.*** A table is **divided into rows** with the **<tr> tag**. Each row is **divided into data cells** with the **<td> tag.** <td> stands for "table data" and holds the content of a data cell.
- **A <td> tag can contain text, links, images, lists, forms, other tables, etc.**
- Tables are made up of rows that contain cells.
  - Cells are the containers for content.



Tables(cont'd)

Table Rows, headers and Cells

- We need two other related tags to make table rows and columns.
  - The **<tr> tag** is used to create a row in the table.
    - Data that will fit in a row will be enclosed in <tr>…</tr> tags.
  - Columns are created by using **<td> tag**.
    - **The number of columns in a table is determined by the number of cells in each row.**
      - For a table with four columns, you need to make sure that every row has **four <td> or <th> elements.**
- **Headers in a table** can be defined with the **<th> tag.**
  - The text in a **<th> element** will be bold and centered.

Example:

```
<table border="1">
  <tr>
      <th>Header 1</th> <th>Header 2</th>
  </tr>
  <tr>
      <td>row 1, cell 1</td>  <td>row 1, cell 2</td>
  </tr>
  <tr>
      <td>row 2, cell 1</td>  <td>row 2, cell 2</td>
  </tr>
  </table>
```

Table tag

Table row

Table head

Table data

Table border

Table Header

Table cell

Header 1	**Header 2**
row 1, cell 1	row 1, cell 2
row 2, cell 1	row 2, cell 2

Attribute	Value	Description
align	Left, Center, right	Specifies the alignment of a table according to surrounding text
bgcolor	rgb(x,x,x), #xxxxxx colorname	Specifies the background color for a table
background	Image url	Sets background image of the table
border	pixels	Specifies the width of the borders around a table
bordercolor	rgb(x,x,x), #xxxxxx colorname	Specifies the color used for the border
cellpadding	pixels	Specifies the space between the cell wall and the cell content
cellspacing	pixels	Specifies the space between cells
width	Pixels | %	Specifies the width of a table
height	Pixels | %	Specifies the height of a table

Tables(cont'd)

Table Cellpadding and Cellspacing:

- The two attributes *cellpadding* and *cellspacing* will allow you to **adjust the white space in your table cell.**
- Cellspacing defines the width of the border, while cellpadding represents the distance between cell borders and the content within.
- Example:

```
<table border="1" cellpadding="15" cellspacing="10">
<tr> <th>Name</th> <th>Salary</th> </tr>
<tr> <td>Ramesh Raman</td> <td>5000</td> </tr>
<tr> <td>Shabbir Hussein</td> <td>7000</td> </tr>
</table>
```

Name	Salary
Ramesh Raman	5000
Shabbir Hussein	7000

Name	Salary
Ramesh Raman	5000
Shabbir Hussein	7000

Tables(cont'd)

Attributes of rows and cells:

- This attributes will be applicable only to the header cell or the data cell if it is used with **<th> or <td> tag**
- It will affect the whole content of the row if it is used by the **<tr> tag.**

Attribute	Value	Description
align	Left | right | center | justify	Aligns the content in a cell
bgcolor	rgb(x,x,x) | #xxxxxx |Colorname	Specifies a background color for a cell
colspan	number	Specifies the number of columns a cell should span
rowspan	number	Sets the number of rows a cell should span
height	Pixels | %	Sets the height of a cell
width	Pixels | %	Specifies the width of a cell
valign	top|middle|bottom	Vertical aligns the content in a table row

Tables(cont'd)

```
<table border="5" bordercolor="green" bgcolor="gray">
<tr>
  <th>Column 1</th>
  <th>Column 2</th>
  <th>Column 3</th>
</tr>
<tr>
<td rowspan="2">Row 1 Cell 1</td>
<td bgcolor="red">Row 1 Cell 2</td>
<td>Row 1 Cell 3</td>
</tr>
<tr><td>Row 2 Cell 2</td><td>Row 2 Cell 3</td></tr>
<tr><td colspan="3">Row 3 Cell 1</td></tr>
</table>
```

Column 1	Column 2	Column 3
Row 1 Cell 1	Row 1 Cell 2	Row 1 Cell 3
	Row 2 Cell 2	Row 2 Cell 3
Row 3 Cell 1		

Tables(cont'd)

❖ Spanning rows and cells

```
<table border="1" cellpadding="2">
    <tr>
        <td>A cell</td>
        <td>Another cell</td>
        <td>Yet another cell!</td>
    </tr>
    <tr>
        <td rowspan="2">A cell that spans two rows</td>
        <td colspan="2">A cell that spans two columns</td>
    </tr>
    <tr>
        <td>Another cell</td>
        <td>The last cell</td>
    </tr>
</table>
```

A cell	Another cell	Yet another cell!
A cell that spans two rows	A cell that spans two columns	
	Another cell	The last cell

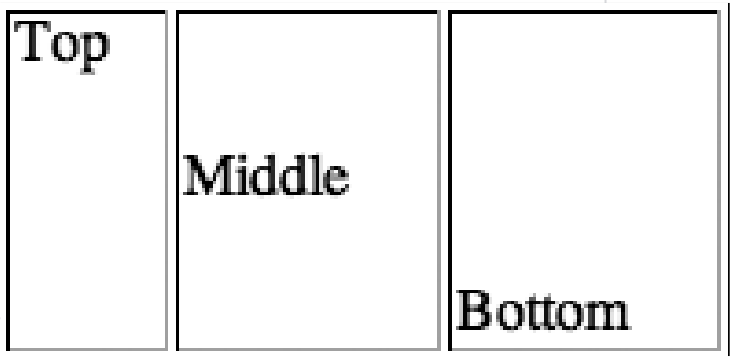
Tables(cont'd)

Vertical alignment of cell content

- If you set your table's width to a small value, web browsers vertically align content in the middle of cells.
- You can use the **valign attribute** to set where the text is displayed vertically.
- The attribute can be added to a **row or cell start tag**, and set to the desired value.

Using a Header, Body, and Footer:

- Tables can be divided into three portions: **a header, a body, and a foot.**
- The three elements for separating the head, body, and foot are:
  - **<thead>** - to create a separate table header.
  - **<tbody>** - to indicate the main body of the table.
  - **<tfoot>** - to create a separate table footer.



Tables(cont'd)

Example:

```
<table border="1" width="100%">
<thead>
<tr><td colspan="4">This is the head of the
    table</td>  </tr>
</thead>
<tbody>
<tr> <td>Cell 1</td> <td>Cell 2</td>
    <td>Cell 3</td> <td>Cell 4</td> </tr>
<tr>
...more rows here containing four cells...
</tr>
</tbody>
<tbody>
<tr>  <td>Cell 1</td> <td>Cell 2</td>
    <td>Cell 3</td> <td>Cell 4</td></tr>
<tr>
...more rows here containing four cells...
</tr>
</tbody>
 <tfoot>
<tr> <td colspan="4">This is the foot of the
    table</td>  </tr>
</tfoot> </table>
```

Tables(cont'd)

❖ This produces the following output:

This is the head of the table			
Cell 1	Cell 2	Cell 3	Cell 4
Cell 1	Cell 2	Cell 3	Cell 4
This is the foot of the table			

Example

Purchase	Location	Date	Evaluation	Cost (€)
Haircut	*Hairdresser*	*12/09*	*Great idea*	*30*
Lasagna	*Restaurant*	*12/09*	*Regrets*	*18*
Shoes	*Shoeshop*	*13/09*	*Big regrets*	*65*
Toothpaste	*Supermarket*	*13/09*	*Good*	*5*
SUM				**118**

Ordered and Unordered List

- HTML Supports several ways of arranging items in lists.
- The most commonly used are: **ordered list (numbered list),** and **unordered list (bulleted list).**

An ordered list:

1. The first list item
2. The second list item
3. The third list item

An unordered list:

- List item
- List item
- List item

Ordered and Unordered List (cont'd)

Unordered List

- An unordered list starts with the ** tag**. Each list item starts with the ** tag**.
  - The list items are marked with **bullets (typically small black circles).**
- The **"type" attribute** can be used to **specifies the style of the bullets**
  - disc, square and circle.
- Example:

```
<ul>
  <li>Banana</li>
  <li>Orange</li>
</ul>
```

How the HTML code above looks in a browser:

- Banana
- Orange

Ordered and Unordered List (cont'd)

Ordered List

- We use an ordered list **when the sequence of the list items** is important.
- A list of instructions is a good example.
- An ordered list starts with the ** tag**. Each list item starts with the ** tag**.
- The list items are **marked with numbers**.

```
Types of fruits are:
<ol>
    <li>Banana</li>
    <li>Orange</li>
    <li>Apple</li>
</ol>
```

How the HTML code above looks in a browser:

Types of fruits are:

1. Banana
2. Orange
3. Apple

Ordered and Unordered List (cont'd)

❖ **The start attribute:**

- If you want a **numbered list to start at a number other than "1,"** you can use the **start attribute to specify another starting number.**

❖ Example

```
<ol start="17">
    <li>Highlight the text with the text tool.</li>
    <li>Select the Character tab.</li>
    <li>Choose a typeface from the pop-up menu.</li>
</ol>
```

❖ The resulting list items would be numbered 17, 18, and 19, consecutively.

Ordered and Unordered List (cont'd)

❖ The <ol> attributes:

Attribute	Value	Description
start	number	Specifies the start point for the list items
type	Specifies which kind of numbering will be used:	
	1	Normal number like 1, 2, 3, …
	I	Upper case Roman number like I, II, III, …
	i	Lower case Roman number like I, ii, iii, …
	A	Capital letter like A, B, C, …
	a	small letter like a, b, c, …

Ordered and Unordered List (cont'd)

Nesting Lists

- One list can be put inside another to create nested list.
- Example:

```
<ol>
<li>Super computer</li>
<li>Mainframe computer</li>
<li>Mini computer</li>
<li>Micro computer
 <ul>
        <li>Desktop computer</li>
        <li>Laptop Computer</li>
        <li>Palmtop computer</li>
 </ul>
</li>
</ol>
```

1. Super computer
2. Mainframe computer
3. Mini computer
4. Micro computer
    - Desktop computer
    - Laptop computer
    - Palmtop computer

Ordered and Unordered List (cont'd)

Definition lists

- Definition (or dictionary) lists are used for lists of terms with their respective definitions.
  - The whole list is marked up as a definition list (<dl>) element.
- Definition lists vary only slightly from other types of lists in that **list items consist of two** parts**: a term and a definition/description**.
  - The **term** is given by the **<dt> element** and is restricted to inline content.
  - The **description** is given with **a <dd> element** that contains block-level content.
- The content of a <dl> is some number of terms (indicated by the <dt> element) and definitions (indicated by the <dd> element).
- Here is an example of a brief definition list :

```
<dl>
<dt>Linotype</dt>
<dd>Line-casting allowed type to be selected, used, then recirculated into the machine automatically. This advance increased the speed of typesetting and printing dramatically.</dd>
<dt>Photocomposition</dt>
<dd>Typefaces are stored on film then projected onto photo-sensitive paper. Lenses adjust the size of the type.</dd>
<dt>Digital type</dt>
<dd><p>Digital typefaces store the outline of the font shape in a format such as Postscript. The outline may be scaled to any size for output.</p>
<p>Postscript emerged as a standard due to its support of graphics and its early support on the Macintosh computer and Apple laser printer.</p>
</dd>
</dl>
```

Linotype

Line-casting allowed type to be selected, used, then recirculated into the machine automatically. This advance increased the speed of typesetting and printing dramatically.

Photocomposition

Typefaces are stored on film then projected onto photo-sensitive paper. Lenses adjust the size of the type.

Digital type

Digital typefaces store the outline of the font shape in a format such as Postscript. The outline may be scaled to any size for output.

Postscript emerged as a standard due to its support of graphics and its early support on the Macintosh computer and Apple laser printer.

HTML Forms

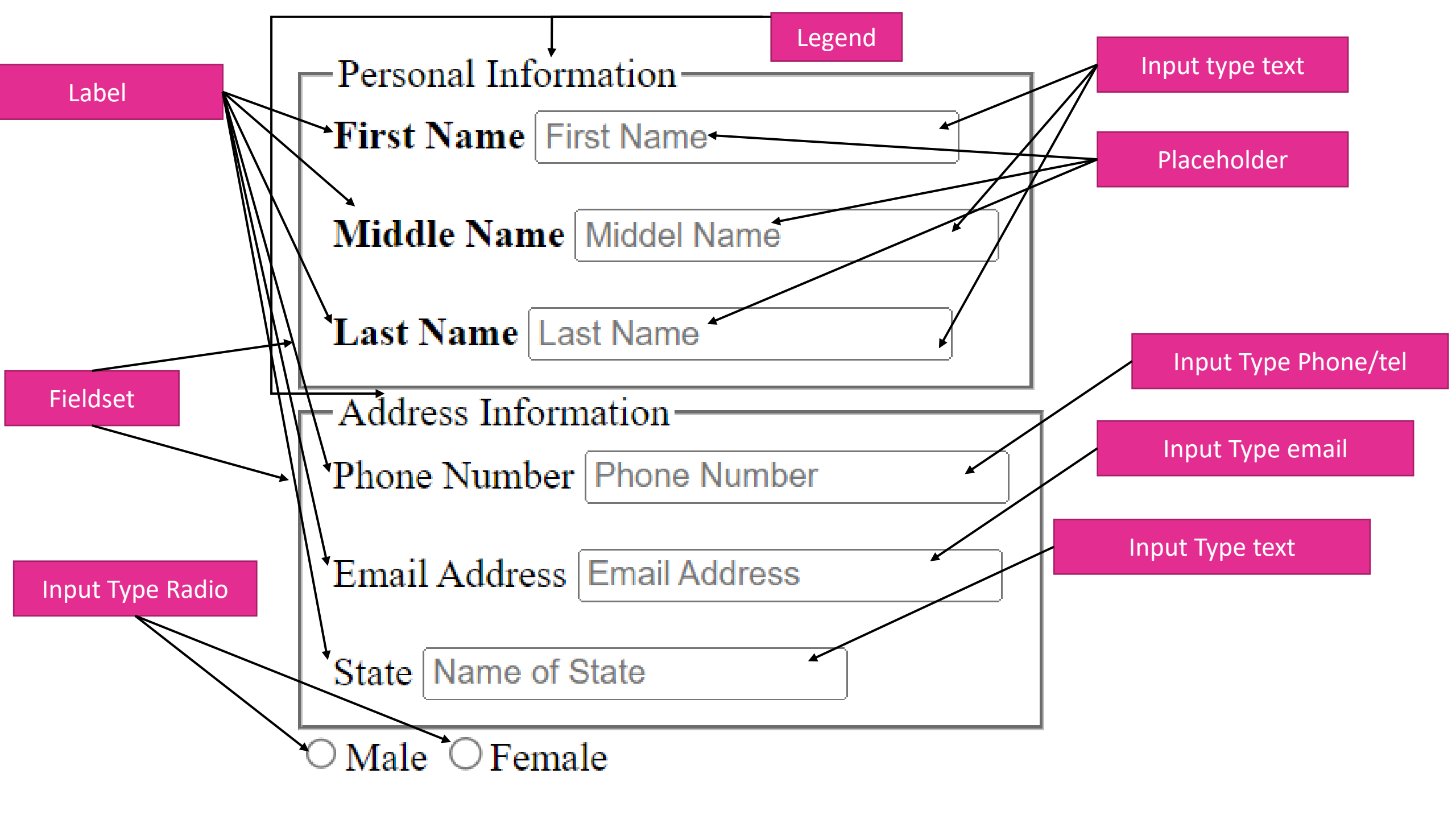
- HTML forms are used to pass data to a server.
  - Users generally fill out the form and submit the form for further processing by the server.
- Forms are a vital tool for the webmaster to receive information from the web surfer, such as their name, email address, credit card, etc.
  - You may store that data in a file, place an order, gather user statistics, register the person to your web forum, or maybe subscribe to your weekly newsletter.
- A form will take input from the site visitor and then will post to your back-end application such as CGI, ASP, PHP script, etc.
  - Then your back-end application will do the required processing on that data in whatever way you like.

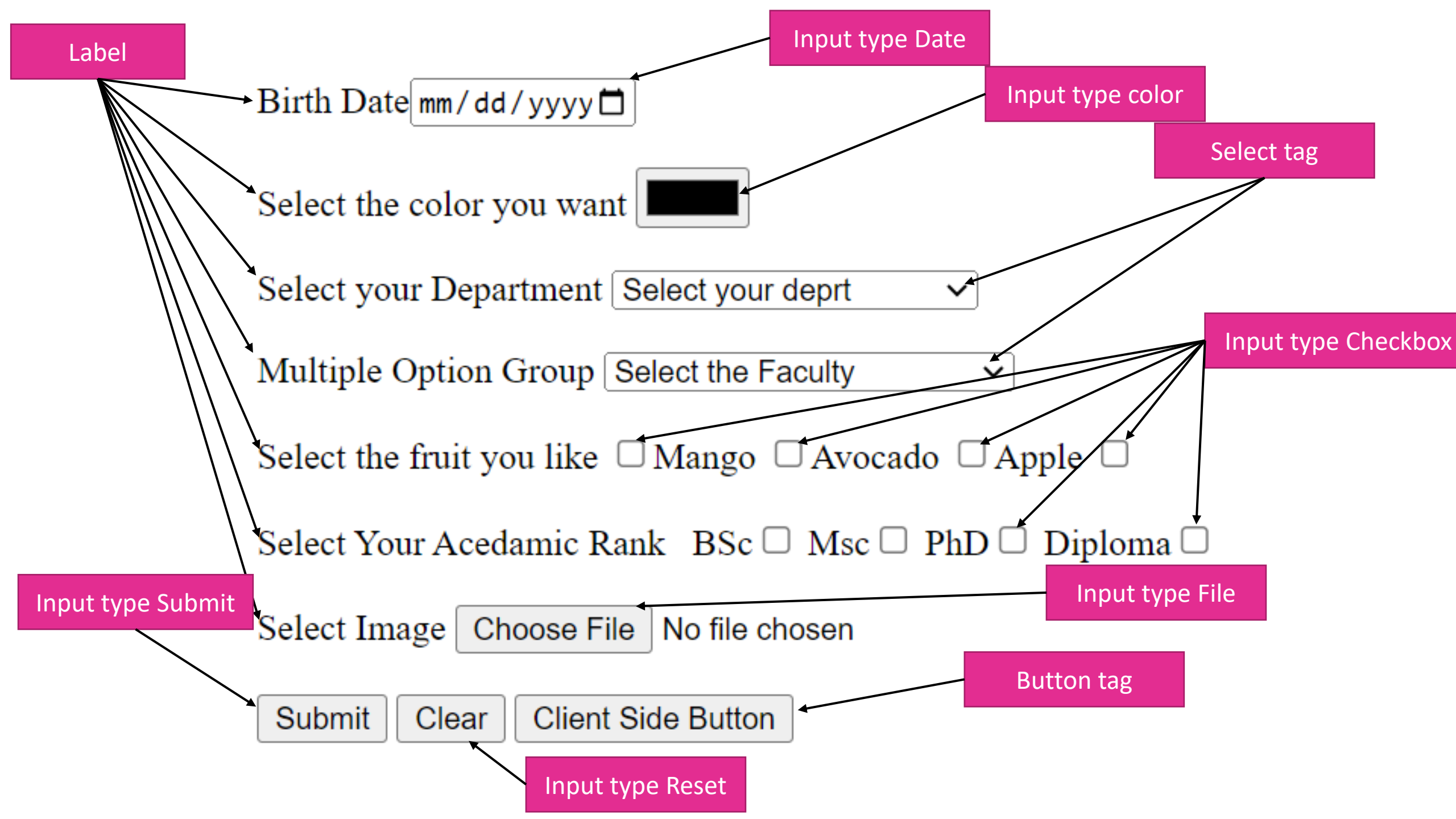
HTML Forms (cont'd)

- The <form> tag is used to create an HTML form:

```
<form>
    input elements
</form>
```

- The most important element inside the form element is the input element.
  - The input element is used to accept user information.
    - An input element can be of type text field, checkbox, password, radio button, submit button, and more.
- The form itself is not visible but form elements are visible.


Legend
Label
Input type text
Placeholder
Personal Information
First Name
First Name
Middle Name
Middel Name
Last Name
Last Name
Fieldset
Input Type Phone/tel
Input Type email
Input Type text
Address Information
Phone Number
Phone Number
Email Address
Email Address
State
Name of State
Input Type Radio
Male
Female


Label
Input type Date
Input type color
Select tag
Input type Checkbox
Input type Submit
Input type File
Button tag
Input type Reset
Birth Date
mm/dd/yyyy
Select the color you want
Select your Department
Select your deprt
Multiple Option Group
Select the Faculty
Select the fruit you like
Mango
Avocado
Apple
Select Your Acedamic Rank
BSc
Msc
PhD
Diploma
Select Image
Choose File
No file chosen
Submit
Clear
Client Side Button

HTML Forms (cont'd)

Text Field

- Text field is a single-line text entry field where the user can enter data.
- You can create a text field by using:

  <input **type="text"** name="name of input field">

  - This defines a one-line input field that a user can enter text into.
- Text fields have two important attributes: **name and value**.
  - The **name attribute** gives a name to the text field for identification purposes and to make it easily accessible.
  - The **value attribute** sets the content of the text field.

```
<form>
  First name: <input type="text" name="firstname"><br />
  Last name: <input type="text" name="lastname">
</form>
```

HTML Forms (cont'd)

- **name** - the **name** attribute is required for identifying the input field name.
- **value** - the **value** attribute specifies default text that appears in the field when the form is loaded.
- **size** - by default, browsers display a text-entry box that is 20 characters wide, but you can change the number of characters using the **size** attribute.
- **maxlength** - by default, users can type an unlimited number of characters in a text field regardless of its size.
- **minlength** - the minimum number of characters long the input can be and still be considered valid
- **readonly** - the text field becomes a read-only input field.
- **disabled** - the text field becomes disabled which means it doesn't accept any input.
- **required** - you can use the required attribute as an easy way of making entering a value mandatory before form submission is allowed.
- **placeholder** - a value to display in the input field whenever it is empty

HTML Forms (cont'd)

❖ Inputting numbers

- Input element of type number is used to let the user enter only a number.
- They include built-in validation to reject non-numerical entries.
- It can be defined as <input **type="number">.**

Attribute	Description
value	A number representing the value entered into the input. You can set a default value for the input by including a number inside the value attribute
max	The maximum value to accept for this input
min	The minimum value to accept for this input
placeholder	An example value to display inside the field when it's empty
Readonly	A Boolean attribute controlling whether or not the value is read-only
step	A stepping interval to use when using up and down arrows to adjust the value, as well as for validation

HTML Forms (cont'd)

- Example:

```
<form>
Quantity:
    <input type="number" name="qty" min="10" max="100“ >
Price:
    <input type="number" name="price" min="10" max="100">
</form>
```

HTML Forms (cont'd)

Password

- A password field works just like a text entry field, except the characters are obscured from view using an asterisk (*) or bullet (•) characters, or another character determined by the browser.
- We can define a password field like:

```
<input type="password">
```

- Example:

```
<form>
        Password: <input type="password" name="pwd">
</form>
```

HTML Forms (cont'd)

Date

- The date input field lets the user enter a date.
  - This could be either with a textbox that validates the input or a special date picker interface.
- The input UI generally varies from browser to browser.
- The resulting value includes the year, month, and day, but not the time.

```
<form>
  <label for="appoint">Choose your preferred appointment date: </label>
   <input type="date" name="appoint" min="2019-04-01" max="2020-04-30">
</form>
```

HTML Forms (cont'd)

❖ Along with the attributes common to all <input> elements, "date" inputs have the following attributes:

Attribute	Description
value	It represents the date entered in the input fields. You can set a default value for the input field by putting a date in the value attribute.
max	The latest acceptable date
min	The earliest acceptable date
step	The stepping interval, when clicking up and down spinner buttons and/or validating the date

HTML Forms (cont'd)

Text Area

- At times, you may want your users to be able enter multiple line of text.
- For these instances, use the **<textarea> element** that is replaced by a multi-line**, scrollable text entry box when displayed by the browser.**
- Example:

```
<textarea name="comment"> Tell us what you feel about our tutorial with 50 words or less. </textarea>
```

Attributes	Description
rows	Specifies the number of lines of text area should display
cols	Specifies the width of the text area measured in number of characters.
spellcheck	Specifies the text area is checked for spelling errors. The possible values are **true, false or default.**

```
<textarea name="comment" rows="5" cols="100"> Tell us what you feel about our tutorial with 50 words or less.
</textarea>
```

Radio Buttons

- Radio buttons are a popular form of interaction.
- You may have seen them on **quizzes, questionnaires**, and other web sites that give the **user a multiple choice question.**
  - Radio buttons let a **user select ONLY ONE** of a limited number of choices.
- We can define a radio button like:

```
<input type="radio">
```

- A radio button works as a group.
  - A radio group is defined by giving each of radio buttons in the group the same name.
    - ➢ Selecting any radio button in that group automatically deselects any currently-selected radio button in the same group.

HTML Forms (cont'd)



- Radio button attributes are:
  - *name*: sets name of the radio button
  - *value*: sets the value of the radio button. This is the data sent to server when the user submits the form.
  - *checked*: sets whether the radio button is checked by default or not. It accepts the value checked.
- Example:

```
<form>
   <input type="radio" name="gender" value="male" checked="checked"> Male <br>
   <input type="radio" name="gender" value="female"> Female

</form>
```

HTML Forms (cont'd)

Checkboxes

- Check boxes allow for **multiple items to be selected** for a certain group of choices.
- The check box's name and value attributes behave the same as a radio button.
- We can define a checkbox like:

```
<input type="checkbox">
```

- Checkboxes let a user select ONE or MORE options.



- Example:

```
<form>
<input type="checkbox" name="vehicle" value="Bike"> I have bike <br>
<input type="checkbox" name="vehicle"  value="Car"> I have car
</form>
```

HTML Forms (cont'd)

Attributes	Description
name	sets name of the checkbox
Value	sets the value of the checkbox. This is the data sent to server when the user submits the form.
Checked	sets whether the checkbox is checked by default or not. It accepts the value checked

❖ Example:

What type of food do you like?

```
<ul>
<li><input type="checkbox" name="genre" value=" spaghetti " checked="checked">
Spaghetti</li>
<li><input type="checkbox" name="genre" value="pizza" checked="checked"> Pizza</li>
<li><input type="checkbox" name="genre" value="sandwich">Sandwich </li>
<li><input type="checkbox" name="genre" value="Burger">Burger</li>
</ul>
```

HTML Forms (cont'd)

Selection lists

- <select> element represents a control that provides a menu from which a user can choose.
  - **Drop down menus** are created with the **<select> and <option> tags.**
- <select> is the list itself and each <option> is an available choice for the user.
- Example:

```
Educational level: <br>
<select name="degree">
<option>Choose One</option>
<option>Some High School</option>
<option>High School Degree</option>
<option>Some College</option>
<option>Bachelor's Degree</option>
<option>Doctorate</option>
</select>
```

HTML Forms (cont'd)

Attributes of select

Attribute	Value	Description
disabled	disabled	Specifies that a drop-down list should be disabled
multiple	multiple	Specifies that multiple options can be selected
name	text	Specifies the name of a drop-down list
size	number	Specifies the number of visible options in a drop-down list

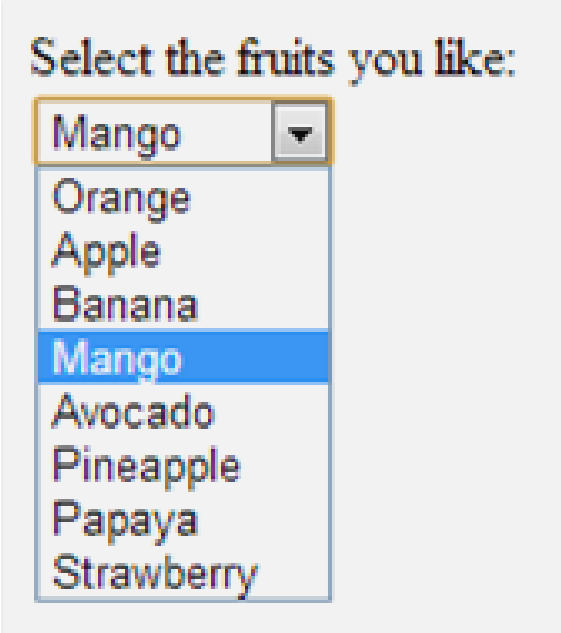
Attribute of option

Attribute	value	Description
selected	selected	Specifies whether the option is selected or not when the form loads

HTML Forms (cont'd)

❖ Example:

```
Select the fruits you like:
<select name="fruits" size="6" multiple="multiple">
    <option>Orange</option>
    <option>Apple</option>
    <option selected="selected">Banana</option>
    <option selected="selected">Mango</option>
    <option> Avocado</option>
    <option>Pineapple</option>
    <option>Papaya</option>
    <option>Strawberry</option>
</select>
```





HTML Forms (cont'd)

Grouping menu options

- You can use **the <optgroup> element** to create conceptual groups of options.
- The required **label attribute** in the optgroup element provides the heading for the group.

```
<select name="icecream" multiple="multiple">
    <optgroup label="traditional">
        <option>vanilla</option>
        <option>chocolate</option>
    </optgroup>
    <optgroup label="fancy">
        <option>Super praline</option>
        <option>Nut surprise</option>
        <option>Candy corn</option>
    </optgroup>
</select>
```

HTML Forms (cont'd)

Hidden controls

- There may be times when you need to send information to the form processing application that does not come from the user.
  - In these instances, you can use a hidden form control that sends data when the form is submitted, but is not visible to user.

```
<input type="hidden">
```

- Hidden controls are added using the input element with the type set to hidden.
  - Its sole purpose is to pass a name/value pair to the server when the form is submitted.
- Example:

```
<input type="hidden" name="isbn " value="1234567890">
```

HTML Forms (cont'd)

File selection control

- Web forms can collect more than just data. They can also be used to transmit external documents from a user's computer.
- The **file selection control** makes it possible for users to **select a document from the hard drive to be submitted with the form data.**
  - It is added to the form using the input element with its type set to file.
- Example:

```
<input type="file“ name=“cv">
```

- The browser displays a "**file" input as a text field with a button** that allows the user to navigate the hard drive and select the file for upload.

HTML Forms (cont'd)

- Example:

```
<input type="file" name="photo" size="28" id="form-photo">
```

- It is important to note that when a form contains a file selection input element, you must specify the encoding type (**enctype)** of the form as **multipart/form-data** and use the POST method.

```
<form method="post" enctype="multipart/form-data">
<p> Send a photo to be used as your online icon:<br>
<input type="file" name="photo" size="28" id="form-photo" />
</p>
</form>
```

HTML Forms (cont'd)

❖ In addition to the common attributes, file inputs support the following attributes:

Attribute	Description
accept	One or more unique file type specifiers describing file types to allow.
capture	What source to use for capturing image or video data
files	A FileList listing the chosen files
multiple	A Boolean which, if present, indicates that the user may choose more than one file

```
<input type="file" accept="image/*,.pdf">
```

HTML Forms (cont'd)

- The accept attribute takes as its value a string containing one or more unique file type specifiers, separated by commas.
- Each unique file type specifier may have the following forms:
  - A valid case-insensitive filename extension, starting with a period (".") character. For example: .jpg, .pdf, or .doc.
  - A valid MIME type string, with no extensions.
  - The string audio/* meaning "any audio file".
  - The string video/* meaning "any video file".
  - The string image/* meaning "any image file".
- For example, a file picker that accepts an image, including both standard image formats and PDF files, might look like this:

```
<input type="file" accept="image/*,.pdf">
```

HTML Forms (cont'd)

Button

- There are a number of different kinds of buttons that can be added to web forms.
- Some are:
  - submit button,
  - reset button, and
  - client side button.

Submit Button

- When clicked, **the submit button immediately sends the collected data** in the form to the server for processing.
- Submit button is defined as follows:

```
<input type="submit">
```

HTML Forms (cont'd)

❖ Submit button has the following attributes:

- value: this sets the text displayed on the button as a label.
- name: used to give name to the submit button

❖ Example:

```
<input type="submit" name="info" value="Send">
```

Reset Button

- The **reset button returns the form controls to the state they were in when the form loaded.**
  - ➢This clears the text users typed into text fields, and removes selections made.
- Reset button can be defined as follows:

```
<input type="reset">
```

HTML Forms (cont'd)

Client Side Button

- This is a button that is used to trigger a client-side script when the user clicks on that button.
  - This is used to execute scripting languages such as JavaScript.
- It has no predefined function on its own, as the submit and reset buttons do.
- Client side button can be defined as follows:

```
<input type="button">
```

- Client side button has the following attributes:
  - value: this sets the text displayed on the button as a label.
  - name: used to give a name to the button
- Example:

```
<input type="button" name="adder" value="Add">
```

HTML Forms (cont'd)

Image Button

- This type of input control allows you to replace the submit button with an image of your choice.
- The image will appear flat, not like a 3-D button.
- Image button can be defined as follows:

```
<input type="image">
```

- Image button attributes:
  - src: sets the image to be used as the submit button
  - value: text displayed on the button
  - name: name of the submit button
- Example:

```
<input type="image" src="didessa.png" value="Submit">
```

HTML Forms (cont'd)

Label

- The <label> element represents a caption for an item in a user interface.
- To associate the <label> with an <input> element, **we need to give an id to the <input> element using *id* attribute.**
  - The <label> then needs a *for* attribute whose value is the same as the input element's id.
- The *for* attribute defines the control the label is associated with.

```
<label for="fname">First name: </label> <br>
<input type="text" name="fname" id="fname">
<label for="lname">Last name: </label> <br>
<input type="text" name="lname" id="lname">
```

```
<label>Do you like peas?
 <input type="checkbox" name="peas">
</label>
```

- Alternatively, you can nest the <input> directly inside the <label>, in which case the for and id attributes are not needed because the association is implicit:

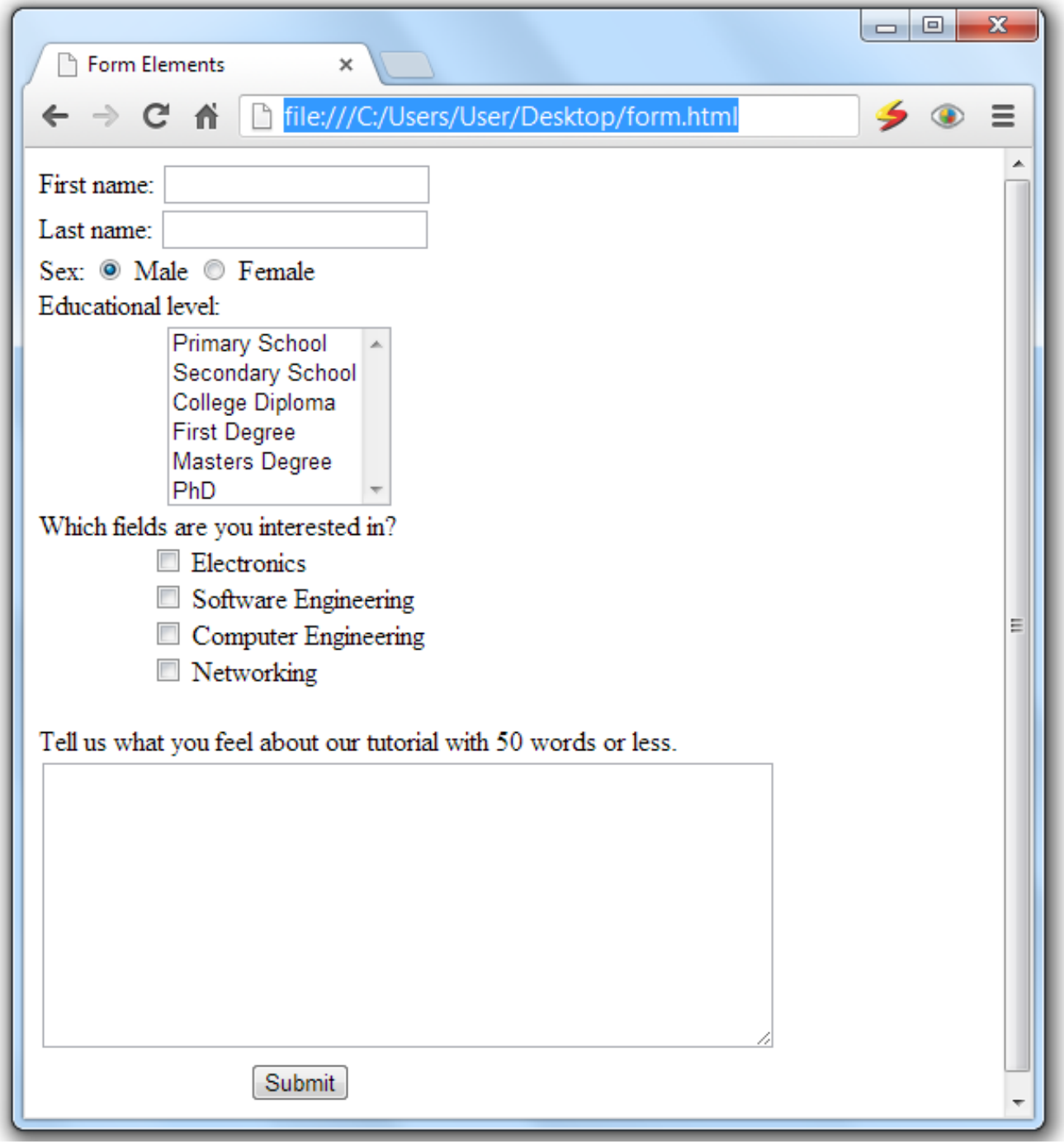
HTML Forms (cont'd)

The fieldset control

- The <fieldset> tag is used to group together a set of related form elements.
  - It is used to group several controls and labels together in a web form.
- The <legend> tag assigns a caption to a fieldset.

```
<form action="/subscribe" method="post">
  <fieldset>
    <legend>Subscribe to the Newsletter</legend>
    <input type="email" name="email">
    <button>Ok</button>
  </fieldset>
</form>
```




Form Elements
file:///C:/Users/User/Desktop/form.html
First name:
Last name:
Sex: Male Female
Educational level:
Primary School
Secondary School
College Diploma
First Degree
Masters Degree
PhD
Which fields are you interested in?
Electronics
Software Engineering
Computer Engineering
Networking
Tell us what you feel about our tutorial with 50 words or less.
Submit

```
<html>
<head> <title>Form Elements</title> </head>
<body>
<form name="test" method="post">
First name: <input type="text" name="fname"> <br>
Last name: <input type="text" name="lname"> <br>
Sex: <input type="radio" name="sex" value="male" checked> Male
      <input type="radio" name="sex" value="female"> Female
<br>
Educational level: <br>
<select name="education" size="6">
          <option>Primary School</option>        <option>Secondary School</option>
          <option>College Diploma</option>      <option>First Degree</option>
           <option>Masters Degree</option>       <option>PhD</option>
</select>  <br>
Which fields are you interested in? <br>
<input type="checkbox" name=“choice1" value="Electronics"> Electronics <br>
<input type="checkbox" name=“choice2" value="Software Engineering"> Software Engineering <br>
<input type="checkbox" name=“choice3" value="Computer Engineering"> Computer Engineering <br>
<input type="checkbox" name=“choice4" value="Networking"> Networking
<br> <br>
Tell us what you feel about our tutorial with 50 words or less. <br>
<textarea name="comment" rows="10" cols="50">  </textarea> <br>
<input type="submit" name="submit">
</form>
</body>
</html>
```

Inserting Multimedia

- HTML allows to insert multimedia data such as **video and audio into web pages.**
- There are two tags for this: <audio> and <video>.
- But, it is not possible to support all kinds of videos and audios on websites.
- Only the most common video and audio formats are supported.

A. Audio

- The **<audio> tag** defines sound, such as music or other audio streams.
  - Currently, there are three supported file formats for the <audio> element: MP3, WAV, and OGG.

Inserting Multimedia (cont'd)

❖ The attributes of <audio> are:

Attribute	Value	Description
autoplay	autoplay	Specifies that the audio will start playing as soon as it is ready
controls	controls	Specifies that audio controls should be displayed (such as a play, pause buttons etc.)
loop	loop	Specifies that the audio will start over again, every time it is finished
muted	muted	Specifies that the audio output should be muted
preload	auto metadata none	Specifies if and how the author thinks the audio should be loaded when the page loads
src	URL	Specifies the URL of the audio file



Inserting Multimedia (cont'd)

- The controls attribute adds audio controls, like play, pause, and volume.
- **Text between the <audio> and </audio> tags will display in browsers that do not support the <audio> element.**
- Example:

```
<audio src="animal.ogg" autoplay>
<p>Your browser doesn't support the given audio format</p>
</audio>
```

B. Video

- The <video> tag is used to insert video into web pages.
- Currently, there are three supported video formats for the <video> element: MP4, WebM, and Ogg.

Inserting Multimedia (cont'd)

Attribute	Value	Description
autoplay	autoplay	Specifies that the video will start playing as soon as it is ready
controls	controls	Specifies that video controls should be displayed (such as a play/pause button etc.)
height	pixels	Sets the height of the video player
loop	loop	Specifies that the video will start over again, every time it is finished
muted	muted	Specifies that the audio output of the video should be muted
poster	URL	Specifies an image to be shown while the video is downloading, or until the user hits the play button
preload	auto metadata none	Specifies if and how the author thinks the video should be loaded when the page loads
src	URL	Specifies the URL of the video file
width	pixels	Sets the width of the video player

```
<video src="rabbit320.webm" controls>
<p>Your browser doesn't support HTML5 video.</p></video>
```

Inserting Multimedia (cont'd)

C. Source

- Since browsers can't support all media types, a browser may fail to play a media if the media is not supported on the browser.
  - A good solution to this is to attach multiple media types so that a browser will play the one it can support.
- This can be done by using <source> element for both <audio> and <video>.
- The <source> tag allows you to specify alternative video/audio files which the browser may choose from, based on its media type or codec support.
- Multiple <source> elements can link to different audio files.
- The browser will use the first recognized format.

Inserting Multimedia (cont'd)

❖ Attributes of <source>:

Attribute	Value	Description
src	*URL*	Specifies the URL of the media file
type	*media_type*	Specifies the media type of the media resource

❖ Example: inserting audio into a web page

```
<audio controls>
  <source src="horse.ogg" type="audio/ogg">
  <source src="horse.mp3" type="audio/mpeg">
  Your browser does not support the audio tag.
</audio>
```

HTML Forms (cont'd)

Linking HTML Forms with PHP Scripts

- To create a link between PHP scripts and HTML forms, we need to use the submit button, and the attribute of the <form tag>, specifically "action" and "method".

The Action Attribute

- The action attribute defines the action to be performed when the form is submitted.

The Method Attribute

- The method attribute specifies the HTTP method to be used when submitting the form data. It is value is GET or POST.

Enctype Attribute

- This attribute specifies how the form data should be encoded when submitting it to the server

Buttons:

- A submit button is used to send form data to a server.
- The data is sent to the page specified in the form's action attribute .
- <form name="input" action="action.php" method="get">

  Username: <input type="text" name="user">

  <input type="submit" value="Submit"> </form>

HTML Forms (cont'd)

Automating Processing: Info Forms and Emails

- You can create a simple form, to enable your visitors to send you comments, questions, product support requests, or whatever you'd like.

```
<form  action='mailto:test@yahoo.com' method="post" enctype="text/plain">
Your Name: <input type="text" size="20" name="VisitorName"> <br><br>
Your Comment: <textarea name="VisitorComment" rows="4" cols="20">
</textarea><br><br>
<input type="submit" value="Email This Form">
</form>
```

Grouping Elements: the & Elements

- The ** and elements** offer a generic mechanism for adding structure to documents.
  - The element defines the **content to be inline** and to be **block-level**
- The div element is used to identify a block-level division of text.
  - The tag defines a division or a section in an HTML document.
- You can use a div like a container around a logical grouping of elements on the page.
- The tag is used to group block-elements to format them with CSS.
- By marking related elements as a div and giving it a descriptive name, you give context to the elements in the grouping.

Grouping Elements: the & Elements

- In this example, a div element is used as a container to create a group

```
<div class="listing">
    <cite>The Complete Manual of Typography</cite>, James Felici. A combination of type history and examples of good and bad type.
</div>
```

- **Here is another common use of a div used to break a page into sections for context, structure, and layout purposes.**
- In this example, a heading and several paragraphs are enclosed in a **div** and identified as the "news" section.

```
<div id="news">
    <h1>New This Week</h1>
    <p>We've been working on...</p>
    <p>And last but not least,... </p>
</div>
```

Grouping Elements: the & Elements

- A span offers all the same benefits as the div element.
- It is used for inline elements that do not introduce line breaks.
  - Because spans are inline elements, **they can only contain text and other inline elements.**
- **In other words, you cannot put block-level elements in a span.**
  - Example:

```
<P><span class="client-title">Client information:</span> </P>
```

- Section tag
- Aside tag
- Article tag

The End !!